

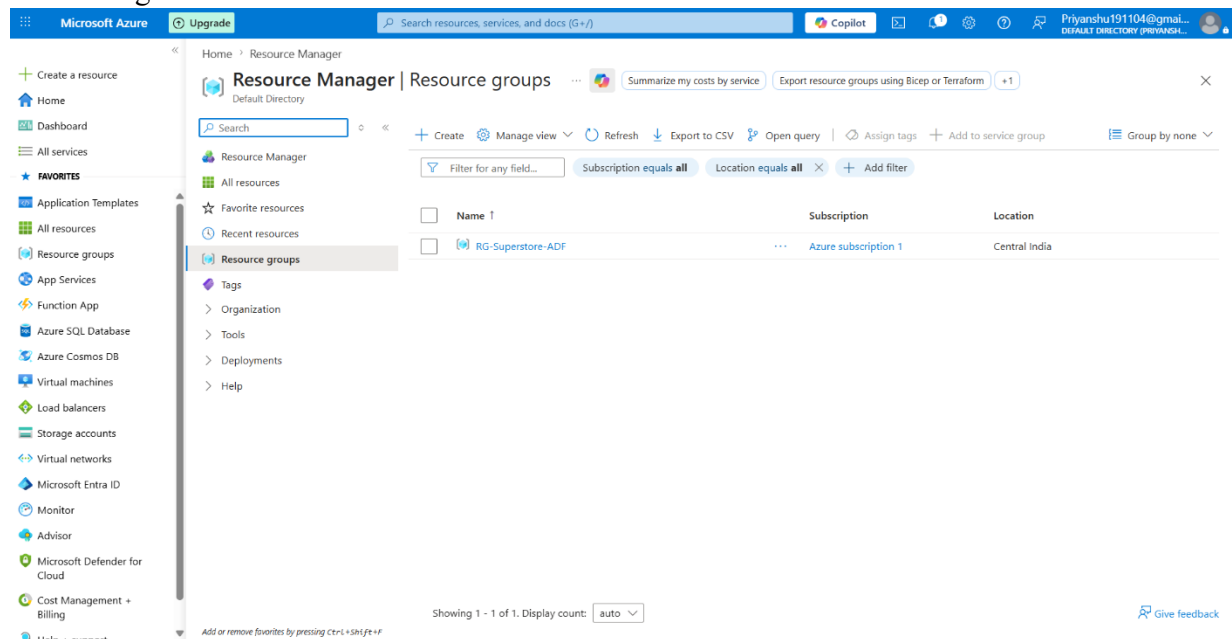
WEEK 4 ASSIGNMENT

INTRODUCTION

This assignment demonstrates the implementation of Azure Blob Storage and Azure Data Factory for copying data from one storage container to another. It also includes IAM role assignment and the use of the Get Metadata activity for retrieving file information before data movement.

TASK 1 – Create Resource Group

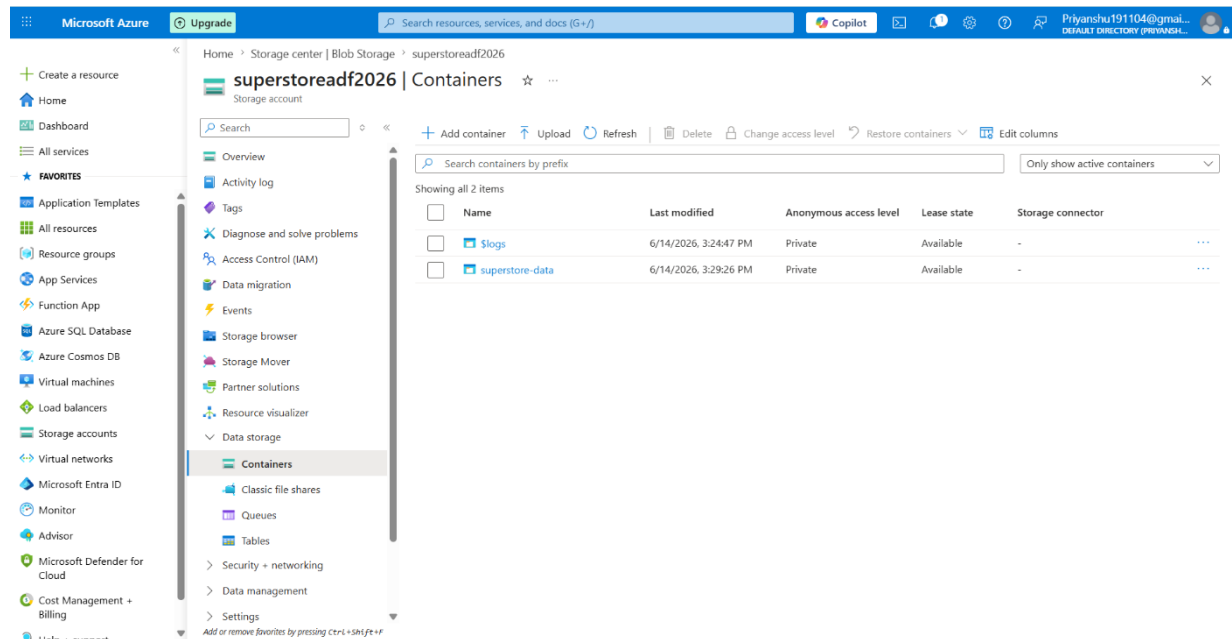
- Resource Group Name: RG-Superstore-ADF
- Region: Central India



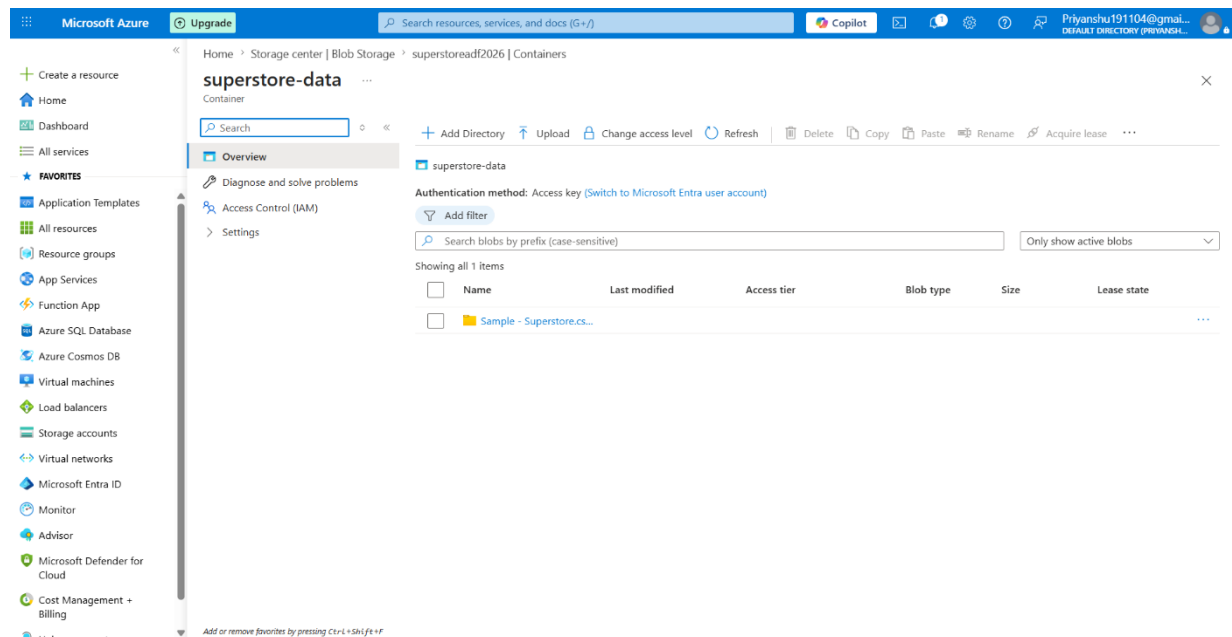
Task 2 – Create Storage Account

Configuration

- Storage Account Name: Superstoreadf2026
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- Region: Central India



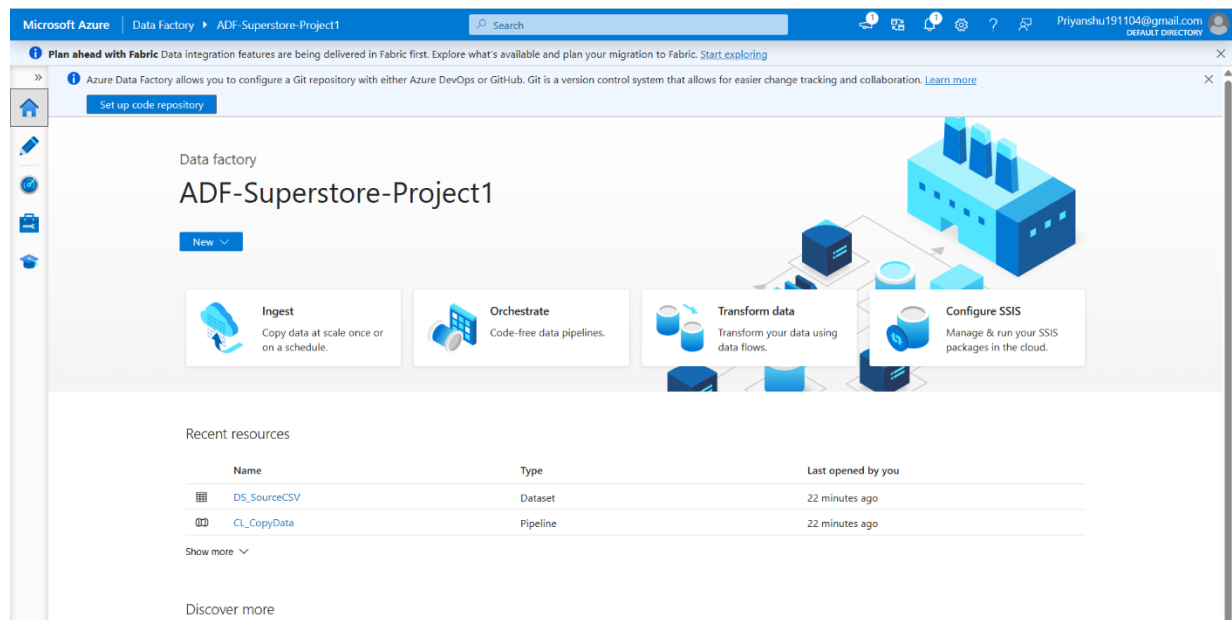
Upload Source CSV File



TASK 3 – ADF Basics

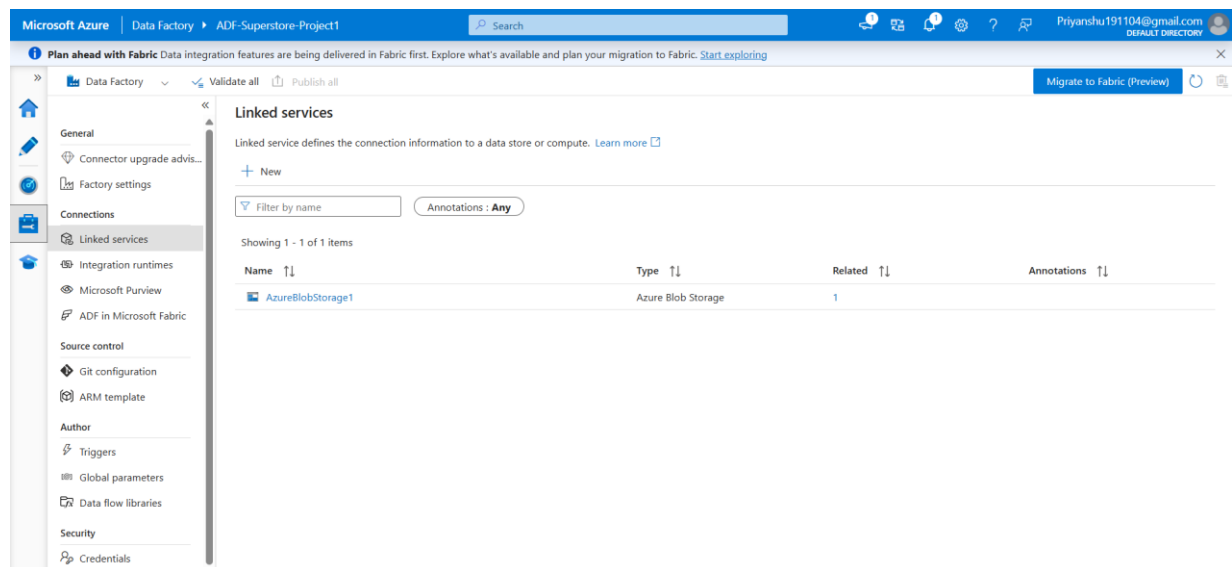
Creation of Azure Data Factory

Azure Data Factory is a cloud-based data integration service provided by Microsoft Azure. It enables users to create, schedule, and manage data pipelines that move and transform data across different storage systems and services.



☐ Create Linked Service

A Linked Service acts as a connection between Azure Data Factory and external data sources. In this assignment, a Linked Service named AzureBlobStorage1 was created to establish a connection between Azure Data Factory and the Azure Blob Storage account. This connection allows the pipeline to securely access files stored in the storage account.



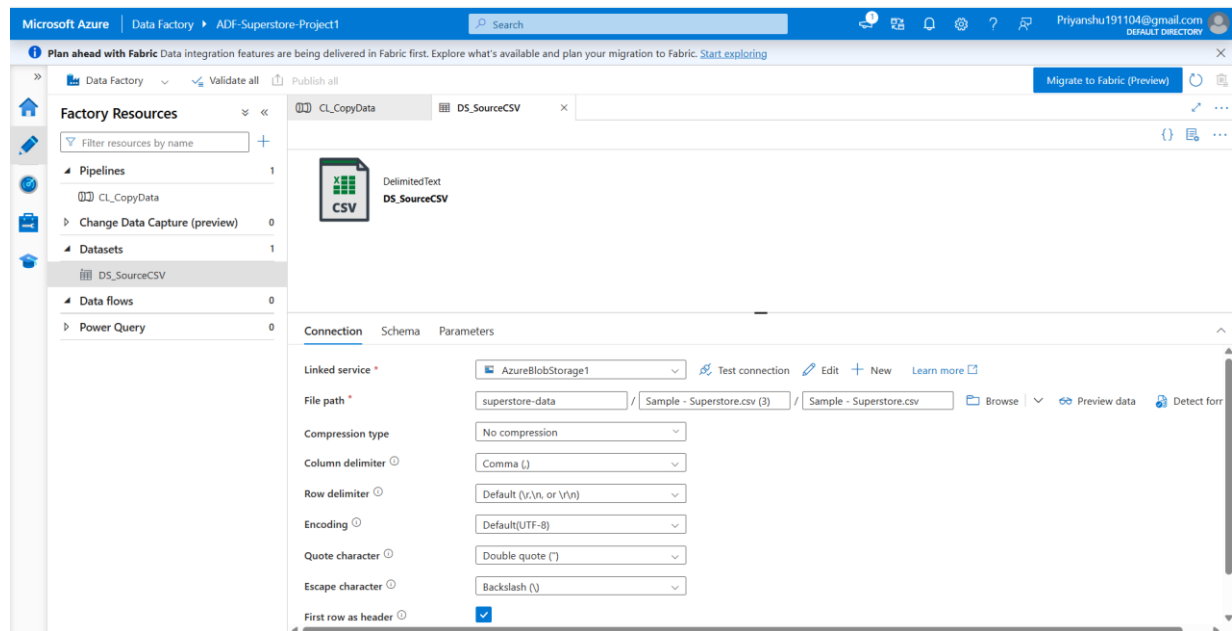
☐ Create Source and Destination Datasets

A Dataset represents the structure and location of data that will be used in a pipeline.

- Name: SourceDataset
- Container: input

- File: Sample - Superstore.csv

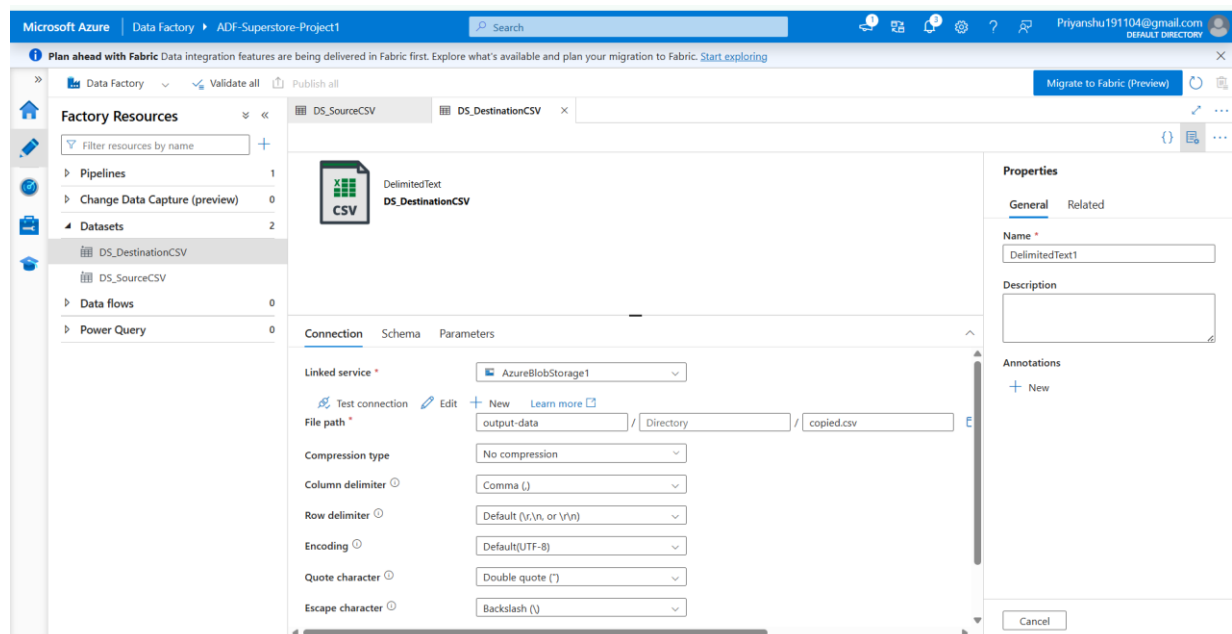
This dataset represents the input file that will be read by the pipeline.



Destination Dataset

- Name: DestinationDataset
- Container: output
- ☐ File: output.csv

This dataset represents the destination where the copied data will be stored.



Use Get Metadata Activity

The Get Metadata activity is used to retrieve metadata information about a file or dataset before performing further operations.

In this assignment, the activity was configured with the SourceDataset and the Exists property was selected to verify whether the source file is available before executing the copy operation.

Purpose of Get Metadata

- Check whether the source file exists.
- Validate the availability of data before processing.
- Improve pipeline reliability by avoiding execution errors.

A

The screenshot displays the Microsoft Azure Data Factory console for a project named 'ADF-Superstore-Project1'. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'CL_CopyData'. The main canvas shows the pipeline diagram with two activities: 'Get Metadata' (labeled 'Get Metadata1') and 'Copy data' (labeled 'Copy data1'), connected by a sequence arrow. Both activities have green checkmarks indicating success. Below the diagram, the 'Output' tab shows the pipeline run details for ID '9414bce1-3688-4f0f-8412-5ba3e49e2aba', with a status of 'Succeeded'. A table lists the activities and their execution details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	6/15/2026, 10:48:00 AM	15s	AutoResolveIntegrationRuntime (Central
Get Metadata1	Succeeded	Get Metadata	6/15/2026, 10:47:42 AM	16s	AutoResolveIntegrationRuntime (Central

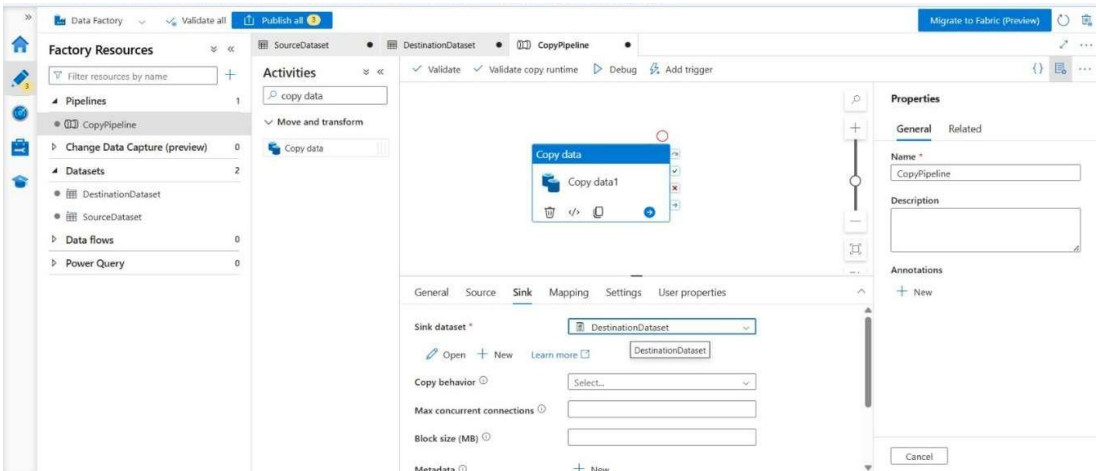
Task 4: Pipeline Development

Objective

The objective of this task is to create an Azure Data Factory pipeline that copies data from a source dataset to a destination dataset using the Copy Data activity.

Pipeline Development

A pipeline named CopyPipeline was created in Azure Data Factory. The Copy Data activity was configured to read data from the SourceDataset (stored in the input container) and copy it to the DestinationDataset (stored in the output container).



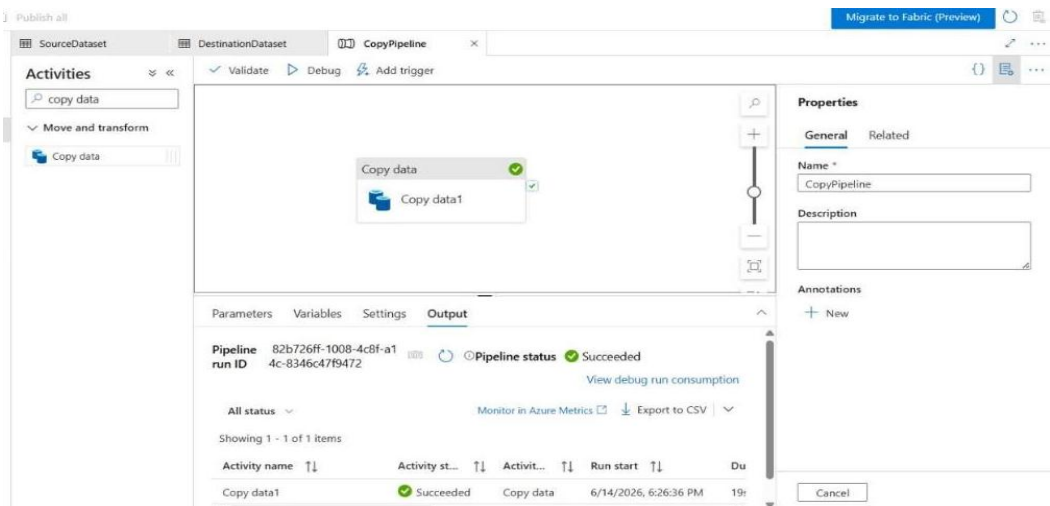
Task 5: Pipeline Execution

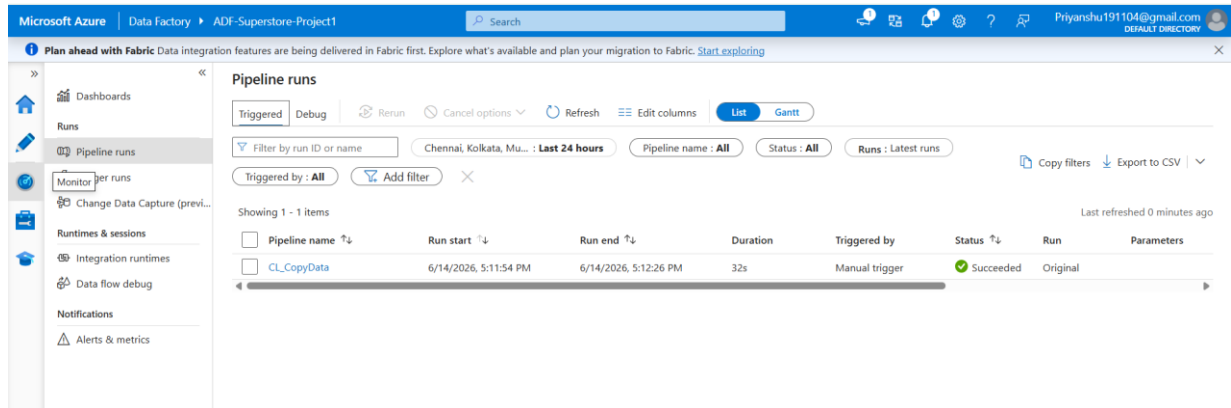
Objective

The objective of this task is to execute the Azure Data Factory pipeline and verify that the data transfer process completes successfully.

Pipeline Execution

The pipeline was executed using the Debug option in Azure Data Factory. During execution, the pipeline copied the data from the SourceDataset in the input container to the DestinationDataset in the output container.





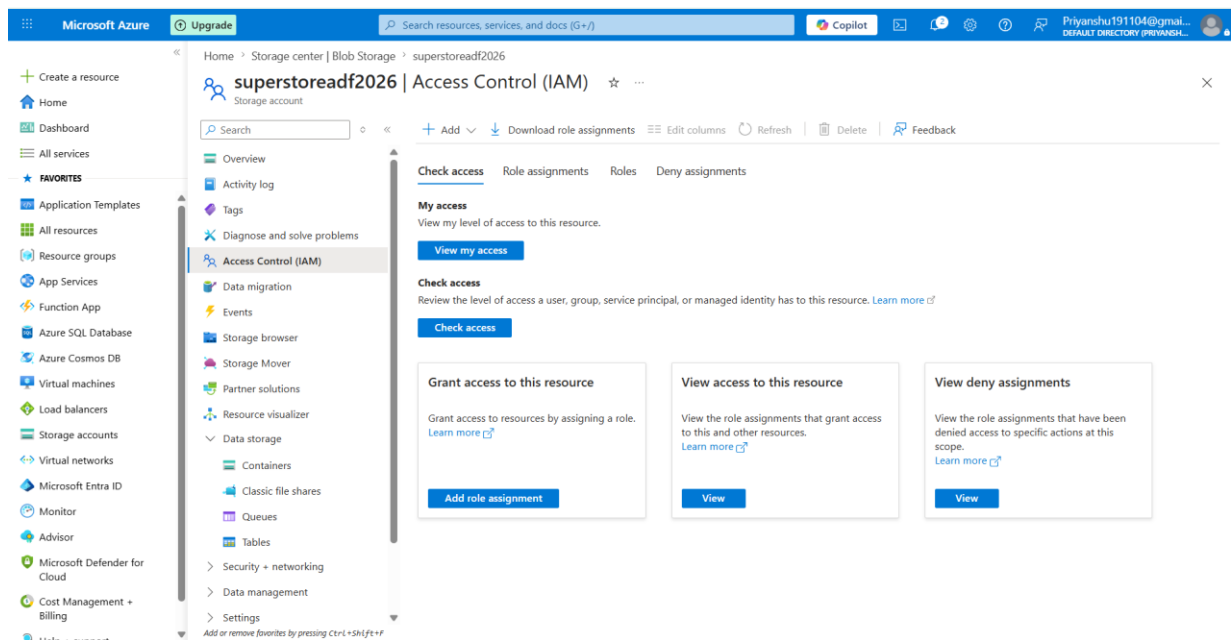
Task 6: IAM Roles

Objective

The objective of this task is to manage access permissions by assigning appropriate IAM (Identity and Access Management) roles to users and Azure Data Factory.

Role Assignment

The Reader role was assigned to provide read-only access to the storage resources. Additionally, the Storage Blob Data Contributor role was assigned to Azure Data Factory to allow it to read and write data in Azure Blob Storage, enabling successful pipeline execution.



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Number of role assignments for this subscription ⓘ

4 4000

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (4) Job function roles (2) Privileged administrator roles (2)

☐	Name ↑↓	Type ↑↓	Role ↑↓	Scope ↑↓	Condition ↑↓
✓	Priyanshu Tiwari MM... 675b4ab6-ed3d-41ac...	User	Owner	Subscription (Inherited)	None
✓	Priyanshu Tiwari MM... 675b4ab6-ed3d-41ac...	User	Contributor	This resource	None
✓	Priyanshu Tiwari MM... 675b4ab6-ed3d-41ac...	User	Reader	This resource	None
✓	ADF-Superstore-Pro... 7a2cc981-f47c-4a15...	Managed identity	Storage Blob Data Contribu...	This resource	Add

Showing 1 - 4 of 4 results.

https://portal.azure.com/?ocid=AIDcmf1el9v5_SEM_k_8e3224508e9315bd14404e7ffad0e24_k_#blade/Microsoft_Azure_AD/RolePermissionsandingBlade/allowRoleSelection/priorityRoles/%5B%5D/roleId/%2Fsubscriptions%2F036c79c-52bf-4e95-ab90-58d8152688f5%2Fprovi...

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Owner ...

Built-in Role

Permissions JSON Assignments

Description: Grants full access to manage all resources, including the ability to assign roles in Azure RBAC.

Search permissions Type: All

Permission Type ☒ Actions ☐ DataActions

Showing 500 of 19940 permissions View all (will take a moment to load)

Type	Permissions	Description
Microsoft.AAD		
Other	Subscription Registration Action ⓘ	Subscription Registration Action
Other	Unregister Domain Service ⓘ	Unregister Domain Service
Other	Register Domain Service ⓘ	Register Domain Service
Read	-- ⓘ	--
Read	-- ⓘ	--
Read	Read Domain Service ⓘ	Read Domain Services
Write	Write Domain Service ⓘ	Write Domain Service
Delete	Delete Domain Service ⓘ	Delete Domain Service
Read	Get the network endpoints of all outbound dependencies ⓘ	Get the network endpoints of all outbound dependencies
Read	Read diagnostic setting for Domain Service ⓘ	Gets the diagnostic setting for Domain Service

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Azure Cosmos DB

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Storage accounts

Virtual networks

Microsoft Entra ID

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Contributor

BuiltInRole

PermissionsJSONAssignments

Description: Grants full access to manage all resources, but does not allow you to assign roles in Azure RBAC, manage assignments in Azure Blueprints, or share image galleries.

Search permissions

Type: All

Permission Type

ActionsDataActions

Showing 500 of 19895 permissions View all (will take a moment to load)

Type	Permissions	Description
Microsoft.AAD		
Other	Subscription Registration Action	Subscription Registration Action
Other	Unregister Domain Service	Unregister Domain Service
Other	Register Domain Service	Register Domain Service
Read	--	--
Read	--	--
Read	Read Domain Service	Read Domain Services
Write	Write Domain Service	Write Domain Service
Delete	Delete Domain Service	Delete Domain Service
Read	Get the network endpoints of all outbound dependencies	Get the network endpoints of all outbound dependencies
Read	Read diagnostic setting for Domain Service	Gets the diagnostic setting for Domain Service

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Reader

BuiltInRole

PermissionsJSONAssignments

Description: View all resources, but does not allow you to make any changes.

Search permissions

Type: All

Permission Type

ActionsDataActions

Showing 500 of 8936 permissions View all (will take a moment to load)

Type	Permissions	Description
Microsoft.AAD		
Read	--	--
Read	--	--
Read	Read Domain Service	Read Domain Services
Read	Get the network endpoints of all outbound dependencies	Get the network endpoints of all outbound dependencies
Read	Read diagnostic setting for Domain Service	Gets the diagnostic setting for Domain Service
Read	Gets the available logs for Domain Service	Gets the available logs for Domain Service
Read	Metrics for Domain Service	Gets metrics for Domain Service
Read	Read Ou Container	Read Ou Containers
microsoft.aadiam		
Read	Azure AD Metrics Definition read	Read Azure AD Metrics Definition

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Storage Blob Data Contributor

Built-in Role

PermissionsJSONAssignments

Description: Allows for read, write and delete access to Azure Storage blob containers and data

Search permissions

Type: All

Permission Type

☒ Actions ☐ DataActions

Showing 6 of 6 permissions

Type	Permissions	Description
Microsoft.Storage		
Other	Generate a user delegation key ⓘ	Returns a user delegation key for the blob service
Write	Patch blob container ⓘ	Returns the result of patch blob container
Delete	Delete blob container ⓘ	Returns the result of deleting a container
Read	Get blob container ⓘ	Returns a container
Read	List of blob containers ⓘ	Returns list of containers
Write	Put blob container ⓘ	Returns the result of put blob container